

SVITLOVODS'K, Ukraine

WMO: 336140

Lat: 49.0711N

Lon: 33.2494E

Elev: 84

StdP: 100.32

Time Zone: 2.00 (EUE)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -17.4	(c) -14.5	(d) -21.2	(e) 0.6	(f) -16.7	(g) -18.5	(h) 0.7	(i) -13.8	(j) 9.9	(k) -1.3	(l) 8.7	(m) -2.6	(n) 2.1	(o) 270	(p) 0.454

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 8.5	(c) 32.1	(d) 20.7	(e) 30.2	(f) 20.6	(g) 28.6	(h) 20.1	(i) 23.2	(j) 28.5	(k) 22.2	(l) 27.7	(m) 21.3	(n) 26.7	(o) 2.7	(p) 100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
21.4	16.2	26.9	20.3	15.1	25.6	19.4	14.3	24.5	69.2	28.4	65.5	27.6	62.2	26.7	28.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 8.1	(b) 6.9	(c) 6.1	(e) -20.5	(f) 34.7	(g) 3.8	(h) 2.1	(i) -23.2	(j) 36.2	(k) -25.4	(l) 37.5	(m) -27.5	(n) 38.7	(o) -30.2	(p) 40.2		
(a) 8.1	(b) 6.9	(c) 6.1	(e) -20.8	(f) 25.2	(g) 3.7	(h) 1.6	(i) -23.5	(j) 26.3	(k) -25.6	(l) 27.2	(m) -27.7	(n) 28.1	(o) -30.3	(p) 29.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.7	-3.8	-2.4	2.4	10.1	16.6	20.6	22.5	21.6	15.9	9.4	3.8	-1.0		
	DBStd	10.25	5.70	5.58	4.39	3.97	3.90	3.19	2.79	3.17	3.61	4.17	4.55	5.18		
	HDD10.0	1656	428	347	237	47	2	0	0	0	2	62	190	342		
	HDD18.3	3515	686	581	494	248	82	14	2	6	89	278	436	600		
	CDD10.0	1549	0	0	1	49	207	319	388	359	179	43	5	0		
	CDD18.3	365	0	0	0	1	28	83	131	107	16	1	0	0		
	CDH23.3	2803	0	0	0	8	221	588	991	873	119	3	0	0		
	CDH26.7	849	0	0	0	1	46	161	317	299	26	0	0	0		
(14) Wind	WSAvg	2.7	3.1	3.2	3.2	2.7	2.4	2.4	2.2	2.3	2.4	2.6	3.2	3.2		
(15) Precipitation	PrecAvg	571	40	32	35	36	51	73	75	55	56	43	40	37		
	PrecMax	830	103	103	86	84	106	279	185	123	144	132	99	82		
	PrecMin	427	12	7	1	0	10	5	7	5	0	0	2	7		
	PrecStd	108	24	22	23	23	27	58	50	34	42	39	25	23		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.6	9.6	16.5	24.6	30.5	32.4	34.1	35.1	29.6	22.7	16.7	11.2		
		MCWB	6.0	6.6	9.6	14.4	18.1	19.7	21.1	20.1	17.8	15.7	13.1	9.6		
	2%	DB	5.5	7.1	13.1	20.9	27.8	30.3	32.0	32.2	26.5	19.6	13.4	8.8		
		MCWB	4.2	5.5	7.8	12.7	17.2	20.2	20.9	21.5	17.3	15.1	11.2	7.6		
	5%	DB	4.0	5.2	10.5	18.7	25.9	28.5	30.1	29.9	24.1	17.2	11.6	6.6		
		MCWB	2.9	3.7	6.3	11.9	17.2	20.0	21.1	20.8	16.9	13.9	10.1	5.6		
	10%	DB	2.8	3.7	8.2	16.7	23.7	26.7	28.3	27.8	22.1	15.3	9.8	5.1		
		MCWB	1.9	2.4	5.3	10.8	16.8	19.6	20.9	20.2	16.1	12.6	8.5	4.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.6	7.5	10.2	15.8	20.8	23.6	24.6	24.9	20.7	17.4	13.5	9.9		
		MCDB	7.3	9.0	15.3	21.4	26.7	28.3	29.9	30.3	25.6	20.3	16.2	11.1		
	2%	WB	4.4	5.6	8.2	13.7	19.5	22.2	23.2	23.1	19.2	15.6	11.7	7.6		
		MCDB	5.4	7.0	12.1	18.8	24.9	27.3	28.5	28.7	23.8	18.5	13.2	8.6		
	5%	WB	3.0	3.9	6.8	12.6	18.3	21.1	22.3	22.0	17.8	14.3	10.2	5.7		
		MCDB	3.8	4.9	9.6	17.1	23.5	26.3	27.8	27.5	22.5	16.7	11.3	6.5		
	10%	WB	2.0	2.6	5.5	11.6	17.2	20.2	21.4	20.9	16.7	13.0	8.7	4.3		
		MCDB	2.7	3.5	8.0	15.6	22.2	25.4	26.8	26.8	20.7	15.1	9.7	5.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.6	5.1	6.1	8.2	8.9	8.2	8.5	9.3	8.2	6.6	4.5	4.1		
		MCDBR	4.1	5.9	10.2	11.6	12.4	11.4	11.7	12.7	12.2	10.0	6.7	5.3		
		MCWBR	3.6	4.6	6.1	5.5	5.1	5.0	4.7	5.3	5.9	5.9	5.1	4.6		
	5% WB	MCDBR	4.0	5.6	8.9	9.6	10.1	9.3	9.5	10.4	10.1	8.3	6.2	5.0		
		MCWBR	3.7	4.7	5.9	5.2	5.3	5.3	5.0	5.7	6.1	5.6	4.9	4.5		
(40) Clear-Sky Solar Irradiance	taub		0.323	0.356	0.387	0.450	0.417	0.410	0.441	0.436	0.409	0.371	0.353	0.326		
	taud		2.194	2.152	2.195	2.086	2.271	2.314	2.234	2.233	2.269	2.352	2.300	2.263		
	Ebn at Noon		701	766	812	795	845	852	817	801	783	752	668	647		
	Edh at Noon		79	105	120	148	128	124	132	127	111	86	73	66		
(44) All-Sky Solar Radiation	RadAvg		0.95	1.77	2.98	4.51	5.83	6.24	6.11	5.47	3.71	2.20	0.91	0.65		
	RadStd		0.18	0.30	0.47	0.41	0.48	0.52	0.34	0.34	0.52	0.32	0.15	0.14		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.82	N/A	N/A	N/A	N/A	N/A	-173	-227	+144	+103
(47) Regional (9 neighbors)		+0.76	N/A	N/A	+1.08	N/A	N/A	-145	-201	+123	+80

Nomenclature: See separate page